
Djezzy National SOC Platform

SS7 Security Module

Security Operations Manual

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Target Audience	SOC Analysts, NOC Engineers, Security Operators

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1. Introduction & Scope

This Security Operations Manual provides standardized procedures for operating the Djezzy SOC Platform's SS7 Security Module. It is designed for frontline security analysts, NOC engineers, and operations staff responsible for day-to-day monitoring, incident response, and maintenance of telecommunications signaling security capabilities. The manual establishes consistent workflows that ensure operational excellence while maintaining compliance with ANRT regulatory requirements.

1.1 Purpose of This Manual

The primary purpose of this manual is to document repeatable procedures that enable operations staff to effectively monitor, analyze, and respond to SS7-specific security threats. By following these standardized procedures, Djezzy ensures consistent response quality regardless of which analyst is on shift, reduces mean-time-to-detect (MTTD) and mean-time-to-respond (MTTR) for signaling attacks, maintains audit trails suitable for regulatory examination, and supports continuous improvement through documented feedback loops.

1.2 Target Audience & Prerequisites

Role	Required Knowledge	Manual Sections
SOC Analyst (SS7 Specialist)	SS7/MAP protocol basics, signaling attack types, threat intelligence	Table of Contents with focus on Ch 4-6
NOC Engineer	Linux administration, Docker, Kafka basics, monitoring tools	Ch 2 & Ch 8 (health checks)
Shift Lead / Supervisor	Incident escalation, team coordination, reporting	Ch 5-7, Ch 10 (escalation)
Security Architect	Full stack knowledge, compliance requirements, capacity planning	Appendixes for review/audit

Training Prerequisite

Before performing independent SS7 monitoring duties, analysts must complete: (1) SS7 Protocol Fundamentals course (16 hours), (2) SS7 Attack Vectors workshop (8 hours), (3) TheHive case management certification (4 hours), (4) Supervised shift operations (40 hours minimum). Contact the Security Training Coordinator for enrollment.

2. Daily Operations Procedures

Daily operations form the foundation of effective SS7 security monitoring. This chapter details the standard operating procedures (SOPs) that must be executed during each shift to maintain visibility into signaling threats, ensure system health, and prepare for potential incidents. Procedures are organized chronologically from shift start to handover completion.

2.1 Shift Start Procedures (First 30 Minutes)

Step 1: System Health Verification (Minutes 0-10)

Begin each shift by verifying all SS7 module components are operational. Access the Grafana SS7 Health dashboard and confirm: (a) All three services show GREEN status (ss7-collector, ss7-analyzer, diameter-monitor), (b) Kafka consumer lag is below 1000 messages for all topics, (c) Elasticsearch cluster status is GREEN with no relocating or initializing shards, (d) TheHive API responds within 500ms latency. Document any anomalies in the shift log before proceeding.

Component	Health Check Command/URL	Expected Result	If Abnormal
ss7-collector	curl -s http://ss7-collector:7000/health	{"status":"healthy","uptime":...}	Check logs, restart if needed
ss7-analyzer	curl -s http://ss7-analyzer:7001/health	{"status":"healthy","rules_loaded":...}	Verify rule files, restart
Kafka clusters	kafka-broker-api --cluster-info	3 brokers, healthy	Check broker logs, network
Elasticsearch	curl -s es-master:9200/_cluster/health?pretty	{ "status" : "green" }	Check shard allocation, disk
TheHive	curl -s -H "X-API-Key: KEY" thehive:9000/api/v1/status	{"status":"OK"}	Check TheHive service

Step 2: Overnight Alert Review (Minutes 10-20)

Review all alerts generated since previous shift handover. In TheHive, filter cases by tag 'ss7-*' created in the last 8 hours. For each CRITICAL or HIGH severity case: (a) Read full case description and observables, (b) Verify assigned analyst has begun investigation, (c) Check if escalation criteria were met and properly executed, (d) Note any cases requiring follow-up during current shift. Update the shift tracking spreadsheet with overnight summary statistics.

Step 3: Baseline Metrics Capture (Minutes 20-30)

Record baseline metrics for comparison throughout the shift. Capture from Grafana dashboards: (a) Current messages per second by type (MAP, ISUP, Diameter), (b) Alert rate (alerts/hour) compared to 7-day average at same time, (c) Top 5 source Global Titles by message volume, (d) Active TheHive case count by severity, (e) Kafka consumer group lag per topic. These baselines enable rapid detection of anomalies developing during the shift.

2.2 Ongoing Monitoring Tasks (Throughout Shift)

Task	Frequency	Tool	Action Threshold
Alert queue monitoring	Every 15 minutes	TheHive UI	>5 unassigned HIGH alerts: immediate triage
Message volume tracking	Hourly	Grafana SS7 Overview	>20% deviation from baseline: investigate
Kafka consumer lag check	Every 30 minutes	Kafka UI or CLI	Lag >10000: notify NOC, consider scaling

Task	Frequency	Tool	Action Threshold
GT blacklist updates	Per intel feed receipt	MISP/OpenCTI	New IOCs: update analyzer rules within 1 hour
PCAP storage utilization	Every 2 hours	df -h /var/capture/ss7	>85%: request storage expansion
Rule effectiveness review	End of each 4-hour shift	ES Query alert/dismissal rate	False >30%: flag for tuning

2.3 Shift End & Handover Procedures

Proper shift handover is critical for maintaining continuous situational awareness. The outgoing analyst must prepare comprehensive documentation enabling seamless transition to incoming personnel. Rushed or incomplete handovers directly contribute to delayed threat detection and response failures.

Handover Item	Format	Location	Completion Criteria
Shift summary narrative	Free text (2-3 paragraphs)	Shift log system	All significant events documented
Active cases list	Table: Case ID, Title, Severity, Status	Shift tracking sheet	All open cases listed with notes
Pending actions	Bullet list with owners/deadlines	Shift tracking sheet	Each action has clear owner and due time
Anomaly observations	Structured notes with timestamps	Shift log + anomaly log	Unexplained patterns flagged for investigation
Metrics snapshot	Key metrics at handover time	Automated dashboard exports	Snapshot saved with timestamp
Tool/system issues	List with workarounds applied	Issue tracker + shift log	Open issues referenced by ticket number

Handover Meeting Requirement

For shifts experiencing CRITICAL alerts or ongoing major incidents, a verbal handover meeting (minimum 15 minutes) is MANDATORY between outgoing and incoming analysts. Schedule via calendar invite with video conference link. Document meeting attendees and key discussion points in the shift log. Skip only when both leads explicitly agree written handover is sufficient.

3. Dashboard Monitoring Guide

Effective monitoring relies on proper interpretation of Grafana dashboards configured for the SS7 module. This chapter explains each dashboard's panels, normal operating ranges, and specific indicators that require analyst attention. Analysts should develop muscle memory for these dashboards through regular practice, enabling rapid anomaly detection without conscious calculation.

3.1 SS7 Security Overview Dashboard

The primary operational dashboard providing real-time visibility into all SS7 module components. This dashboard should be displayed prominently in the SOC operations center and checked minimum every 15 minutes during normal operations, more frequently during active incidents.

Panel Name	Position	Data Source	Normal Range	Warning Indicator
Messages/sec - Total	Top-left, large	Elasticsearch ss7-raw-events	35K-50K EPS	<25K or >60K EPS
Messages/sec by Protocol	Below total	Elasticsearch ss7-raw-events	MAP 55%, ISUP 25%, Diameter 15%, Other 5%	Any protocol >10% deviation
Alert Timeline (24h)	Top-right	Elasticsearch ss7-alerts	Steady low baseline with occasional spikes	Sustained spikes >2x baseline
Top 10 Source GTs	Middle-left	Elasticsearch ss7-raw-events	Known internal GTs dominant	Unknown/unexpected GT in top 10
Active Cases by Severity	Middle-center	TheHive API	Mostly LOW/MEDIUM	HIGH/CRITICAL increasing trend
Geographic Threat Map	Middle-right	Elasticsearch geo_point	Concentrated in Algeria (home)	Unexpected international hotspots
Detection Rule Effectiveness	Bottom-left	Prometheus custom metrics	FP rate <20% for all rules	Any rule FP rate >30%
System Resource Usage	Bottom-right	Node exporter / Docker	CPU <70%, Memory <80%	Any resource >85% sustained

3.2 IRSF Fraud Detection Dashboard

Specialized dashboard focused on International Revenue Share Fraud detection and prevention. This dashboard aggregates financial risk indicators and blocking rule effectiveness metrics. Review this dashboard at minimum twice per shift (mid-shift and near end) even when no fraud indicators are present, as early detection of emerging patterns enables proactive blocking before significant revenue loss occurs.

Panel Name	Purpose	Key Metric	Normal State	Action Trigger
High-Risk Destination Ranking	Identify premium-rate targets	Top countries by call volume	No premium destinations in top 20	Top premium destination appears in ranking
Call Duration Distribution	Detect revenue-generation anomalies	Avg call duration histogram	Peak at 120-300 seconds	Secondary peak <10 seconds (IRSF indicator)
Revenue at Risk (24h)	Quantify financial exposure	\$ amount based on rates	<\$1,000/day estimated	>\$5,000/day: immediate investigation
Blocked GT Count	Measure response effectiveness	Currently blocked source GTs	Stable or slowly increasing	Sharp increase suggests campaign
Blocking Rule Hit Rate	Evaluate rule precision	% of calls matching blocked GTs	>90% legitimate blocks	<80%: possible over-blocking, review rules
Time-to-Block Metric	Measure response speed	Avg seconds from detection to block	<300 seconds (5 min)	>900 seconds: process bottleneck

3.3 Interpreting Anomalous Indicators

Not all dashboard deviations indicate security threats. Many variations reflect normal business cycles such as promotional campaigns, holiday traffic patterns, or scheduled maintenance windows. Analysts must develop contextual judgment to distinguish benign anomalies from genuine threats. When in doubt, investigate rather than ignore - the cost of a false positive investigation is far lower than a missed attack detection.

Observation	Possible Benign Cause	Possible Threat Indication	Investigation Action
Sudden MAP volume spike 09:00	Morning registration peak	Location tracking campaign launch	Check SRI4SM vs normal registration ratio
ISUP increase to country X	New roaming agreement active	IRSF test probe or active fraud	Cross-reference with destination blacklist
Diameter S6a errors increase	HSS maintenance window	IMSI catcher or fake base station	Verify HSS maintenance schedule, check geo
New Global Title in top sources	New STP or interconnect partner	Compromised signaling point	Query GT against known partner list
Alert FP rate increase overnight	Rule threshold needs recalibration	Attack pattern evolution evading detection	Review recent rule changes, sample alerts

4. Incident Response Runbooks

This chapter provides detailed runbooks for responding to SS7-specific security incidents. Each runbook follows a standardized structure: trigger conditions, initial assessment steps, investigation procedure, containment actions, resolution criteria, and post-incident requirements. Analysts should familiarize themselves with all runbooks before an incident occurs, as real-world scenarios leave no time for reading documentation.

4.1 Runbook: Location Tracking Detection (SS7-LocTrack-001)

Trigger Condition:

Alert rule SS7_LOCATION_TRACKING_SUSPICIOUS fires when a single MSISDN receives >10 SRI4SM (Send Routing Info For SM) requests within a 60-second window from the same or related source Global Title(s). Severity defaults to HIGH; escalates to CRITICAL if target subscriber is high-value (VIP, corporate account, government) or if source GT is untrusted/unknown.

Initial Assessment (T+0 to T+5 minutes):

Step	Action	Tool/Command	Expected Output	Decision Point
1	Open TheHive alert, review observables	TheHive Web UI	Source GT, target MSISDN, time of attack	Is target VIP/government? -> Escalate
2	Query source GT reputation	MISP/OpenCTI lookup	IOC match or clean	Known bad GT? -> Skip to containment
3	Check target subscriber profile	CRM/HLR query (authorized)	Subscriber type, account value	Normal subscriber? -> Standard flow
4	Review historical SRI pattern for target	ES query last 30 days	Prior location queries count	History of similar queries? -> Pattern analysis
5	Assess source GT legitimacy	Network topology database	Known partner/interconnect GT	Unknown GT? -> Elevated suspicion

Containment Actions (T+5 to T+30 minutes):

If assessment confirms likely tracking attack: (1) Create P1/P2 TheHive case with all observables attached, (2) Notify Privacy Officer via PagerDuty for subscriber notification decision, (3) Prepare source GT blocking recommendation for network security team (do NOT block without approval), (4) Begin PCAP extraction for evidence preservation with hash integrity verification, (5) If target is VIP, immediately escalate to SOC Manager and Legal liaison.

Resolution Criteria:

Case may be closed when: (a) Source GT is blocked or rate-limited, (b) Target subscriber (if affected) has been notified per legal guidance, (c) Root cause analysis documents how attacker obtained SS7 access, (d) Preventive controls implemented to prevent recurrence, (e) Post-incident review completed with lessons learned captured. Typical resolution time: 4-72 hours depending on complexity.

4.2 Runbook: IRSF Fraud Campaign (SS7-IrsfFraud-002)

Trigger Condition:

Alert rule SS7_IRSF_PATTERN_DETECTED fires when ISUP Initial Address Messages (IAM) show pattern of: (a) Calls to high-risk premium-rate destinations (configurable country list), (b) Short call durations <10 seconds characteristic of revenue generation, (c) Volume >100 calls within 5 minutes from same calling number range, (d) Calling numbers show premium-rate or recently-ported characteristics. Severity is always CRITICAL due to direct financial impact.

Phase	Timeline	Actions	Responsible	Deliverable
Detection	T+0	Alert fires, auto-case created, initial data gathered	SS7 Analyzer (auto)	TheHive case with observables
Triage	T+0 to T+15min	Validate IRSF signature, estimate exposure, identify source	SOC Analyst	Triage assessment document
Containment	T+15min to T+1hr	Block destination ranges, isolate source if internal, preserve CDRs	Network Ops + Fraud Team	Blocking rules applied, CDRs secured
Investigation	T+1hr to T+24hr	Trace attack path, calculate actual loss, identify accomplices	Finance Team + Legal	Investigation report with findings
Recovery	T+24hr to T+72hr	Reverse unauthorized changes, recover lost revenue if possible	Finance + Network Ops	Recovery action log
Closure	T+72hr+	Document lessons learned, update controls, close case	SOC Lead	Post-incident report, updated rules

Financial Impact Documentation

For all IRSF incidents, maintain precise records of: (a) Number of fraudulent calls detected, (b) Average call duration and destination rate, (c) Total estimated revenue loss, (d) Revenue saved by blocking actions, (e) Costs incurred for investigation and recovery. These figures are required for insurance claims, regulatory reports, and law enforcement referrals.

4.3 Runbook: Signaling Denial-of-Service (SS7-SigDoS-003)

Trigger Condition:

Signaling DoS is indicated by: (a) Message flood exceeding 3x baseline EPS from single source GT, (b) Error responses (abort, release) flooding back to originators, (c) STP CPU/memory utilization spiking on nodes receiving traffic, (d) Legitimate call setup failures increasing in correlation with flood timing. This is CRITICAL severity as it impacts network availability for all subscribers.

Priority	Action	Details	Authority Level
1 (Immediate)	Activate traffic filtering at STP edge	Work with Network Ops to apply ACLs limiting suspicious GTs	Network Manager + SOC Manager
2 (Immediate)	Scale SS7 collector/analyser capacity	Add consumer instances, increase partition counts	SOC Engineer (auto-scale if c
3 (Urgent)	Preserve all PCAPs during attack window	Ensure capture continues, verify storage sufficient	NOC Engineer
4 (Urgent)	Notify carrier peers if attack originates externally	Use established GSMA security contact channels	Security Liaison
5 (Follow-up)	Post-attack forensics	Analyze attack source, method, motivation for attribution	Intel Team + Security Architect

5. Investigation Procedures

Beyond structured runbooks for common incident types, analysts frequently encounter unusual patterns requiring open-ended investigation methodology. This chapter establishes the investigative framework for analyzing SS7 security events, including data sources, analytical techniques, evidence handling, and reporting standards applicable across all investigation types.

5.1 Investigation Data Sources

Data Source	Content	Access Method	Retention	Primary Use Case
Raw PCAP files	Complete packet capture with payloads	ads/capture/ss7/ (collector node)	30 days hot, 7 years cold	Deep packet inspection, court evidence
Elasticsearch ss7-raw-events	Normalized JSON messages	Kibana/ES Query API	7 days searchable, 90 days indexed	System search, aggregation, statistics
Elasticsearch ss7-alerts	Generated alerts with context	Kibana/ES Query API	30 days searchable	Alert disposition, trending analysis
TheHive cases	Investigation records with artifacts	TheHive Web UI / API	Permanent	Case management, workflow tracking
Kafka topic replay	Original event stream (if retained)	kafka-console-consumer --from-topics (configurable)	7 days (configurable)	Re-analysis with new rules, debugging
CDR (Call Detail Records)	Billing records from switches	Billing system API (authorized access)	90 days minimum	Correlation with fraud, revenue impact
HLR/VLR logs	Subscriber database queries	Core network logs (via NetOps)	Varies (typically 30 days)	Subscriber targeting verification

5.2 Analytical Techniques

Effective SS7 investigation combines multiple analytical approaches to build comprehensive understanding of threats. No single technique suffices for complex attacks; analysts should systematically apply multiple methods to cross-validate findings and avoid cognitive biases that lead to incorrect conclusions.

Technique	Description	Application Example	Tools Required
Temporal Analysis	Examine timing patterns, intervals, correlations between events	Identify automated vs manual attacks, detect campaign phases	Python (pandas/matplotlib)
Spatial/Geo Analysis	Map source/destination locations, identify geographic clusters	Geographic mapping (subscriber locations), map QoS for origin analysis	GeoIP libraries, map tools
Behavioral Profiling	Compare current behavior against established baselines for entities	Identify anomalous accounts deviating from group usage statistics	ML scoring, group usage analytics
Graph Analysis	Build relationship networks between GTs, MSISDNs, IMSIs	Visualize designated attack infrastructure, identify potential pivots	Neo4j (optional), manual visualization tools
Protocol State Machine	Trace MAP/ISUP dialog state transitions for protocol anomalies	Detect malformed messages indicating wiretaps or SS7 signaling attacks	Wireshark, SS7 protocol analyzers
Correlation Analysis	Link SS7 events with IP/network security events	Connect signaling attack to initial compromise	SIEM cross-domain search, timeline visualization

5.3 Evidence Handling Procedures

SS7 investigations frequently produce evidence subject to legal proceedings, regulatory inquiries, or internal disciplinary actions. Proper evidence handling from moment of detection ensures admissibility and prevents challenges to investigation integrity. Follow chain-of-custody procedures rigorously for all incidents with potential legal ramifications.

Procedure Step	Requirement	Implementation	Verification
Immediate Preservation	Secure raw data before any modification	Copy PCAPs to evidence store, export ESI	Disclosure with (SHA-256) hashes
Access Logging	Record who accessed evidence and when	Evidence management system access logs	Review during case closure
Integrity Verification	Ensure evidence not tampered	Periodic hash recomparison, write-once storage	Hash verification triggers incident
Secure Storage	Encrypt evidence at rest, restrict access	AES-256 encryption, role-based access control	Access attempt logging
Documentation	Maintain clear provenance record	Evidence custody form with transfer signatures	Manager sign-off on transfers

Legal Hold Requirements

When any investigation potentially involves: (a) fraud exceeding \$10,000, (b) government officials or regulated entities as victims, (c) indications of organized crime involvement, or (d) employee misconduct - immediately issue legal hold preserving ALL related data regardless of apparent relevance. Consult Legal before deleting any data in connected systems.

6. Escalation Matrix & Communication

Not all SS7 security events can or should be handled at the analyst level. This chapter defines the escalation framework specifying when and how to involve higher-level resources, external teams, and management. Proper escalation prevents both under-reaction (missing serious threats) and over-reaction (wasting leadership time on routine issues). When in doubt, escalate rather than suppress.

6.1 Severity-Based Escalation Triggers

Current Severity	Escalation Condition	Escalate To	Response SLA	Notification Method
CRITICAL	Always (auto-escalate on creation)	SOC Manager + On-call CISO	15 minute acknowledge	PagerDuty P1 + Phone call + Slack @ch
CRITICAL	No response from primary after 10 min	Deputy SOC Manager + NOC Manager	Immediate	Phone call + PagerDuty escalation
HIGH	Affects VIP/government subscriber	Privacy Officer + SOC Manager	30 minute acknowledge	PagerDuty P2 + Email + Slack
HIGH	Indicates possible insider involvement	HR Security + Internal Audit	1 hour acknowledge	Email to security-hr mailing list
HIGH	External media/press awareness possible	Corporate Communications + Legal	Immediate	Phone call to comms on-call
MEDIUM	Pattern suggests larger campaign	Intel Team for threat landscape review	4 hour acknowledge	TheHive case assignment + Email
MEDIUM	Requires network configuration change	Network Security Team (firewall/SNMP)	Next business day (or earlier if critical)	ServiceNow/Jira + Email
LOW	Recurring false positive from rule	Rules Engineering Team	Weekly batch review	Jira ticket for rule tuning

6.2 External Communication Protocols

SS7 incidents frequently require communication with external parties including peer carriers, law enforcement, regulatory bodies, and vendors. All external communication must follow established protocols to protect Djazzy's interests while maintaining cooperative relationships essential for telecom security. Never admit fault or liability in external communications without Legal review.

Party	Communication Channel	Authorized Content	Approval Required	Template Available
Peer Carrier (abuse complaint)	CSMA ISG portal + email	Technical details of abuse from their network	SOC Manager	Yes - template ABUSE-NOTIF-001
Peer carrier (evidence request)	Official legal channel only	Only with court order or MLAT	Legal Department + CISO	Yes - template EVIDENCE-RESP-001
ANRT (mandatory report)	ANRT secure portal API	Incident summary, affected subscribers	Compliance Officer	Yes - template ANRT-REPORT-001
ANRT (critical incident)	ANRT secure portal + phone	Immediate notification per Circular 05/2012	CISO directly	Yes - template ANRT-CRITICAL-001
Law Enforcement (formal request)	Registered legal address only	Only via formal legal process	Legal Department exclusively	Yes - Legal handles entirely
Vendor (product defect)	Support portal + account manager	Technical details, reproduction steps	Procurement + Engineering	No - case-by-case
Press/Media (inquiry)	Corporate Communications email	Do not comment, never comment	Never respond directly	No - refer to PR immediately

Information Classification Reminder

When communicating externally, classify information appropriately: (INTERNAL) for operational details, (CONFIDENTIAL) for subscriber-identifiable data and investigation methods, (RESTRICTED) for vulnerability specifics and network architecture details. When uncertain, classify one level higher than seems necessary. Misclassification consequences favor over-classification.

7. Maintenance Procedures

Regular maintenance ensures the SS7 module continues operating effectively over time. This chapter documents scheduled maintenance tasks, rule update procedures, model retraining workflows, and capacity planning activities. Maintenance windows should be coordinated with NOC to minimize impact on monitoring coverage during planned work.

7.1 Scheduled Maintenance Calendar

Task	Frequency	Duration	Window	Owner	Impact
PCAP rotation & archival	Daily (02:00)	2 hours	Low traffic (02:00-04:00)	NOC Engineer	None (automated, redundant capture)
Kafka log compaction	Daily (04:00)	1 hour	Low traffic (04:00-05:00)	Kafka Admin	Brief increased latency
Elasticsearch index optimization	Weekly (Sunday 03:00)	4 hours	Maintenance window	ES Admin	Reduced search performance
Detection rule review & tuning	Weekly (Tuesday 10:00)	2 hours	Business hours	Rules Engineer	None (rules versioned, rollback available)
Threat intelligence feed update	Daily (06:00, 14:00, 22:00)	30 min each	Automated + validation	Intel Analyst	Brief rule reload
ML model retraining	Monthly (1st of month)	8 hours	Coordinated maintenance	ML Engineer	Analyzer restart required
Backup verification drill	Monthly (15th)	4 hours	Flexible	SOC Engineer + Backup Admin	Read-only verification
Capacity planning review	Quarterly	4 hours	Business hours	SOC Architect + Capacity Planner	Planning only
Disaster recovery test	Bi-annual	8 hours	Weekend maintenance	Full SOC team + Network Ops	DR Ops active, brief coverage gap

7.2 Detection Rule Update Procedure

Detection rules require regular updates to address new attack variants, reduce false positive rates, and incorporate threat intelligence from MISP/OpenCTI feeds. All rule changes follow a controlled process with testing before production deployment to prevent unintended alert floods or detection gaps.

Phase	Steps	Validation	Rollback Plan
Development	1. Create rule branch in git repo 2. Write/update YAML rule definition 3. Add unit tests for expected matches 4. Test against historical PCAP corpus 5. Document rule logic and thresholds	Unit tests pass False positive rate <5% on test data No performance regression (>10% latency)	Git revert to previous commit Reload old rules from backup
Staging	1. Deploy to staging environment (separate Kafka cluster) 2. Run against live traffic mirror (read-only) 3. Monitor alert output for 24 hours 4. Get peer analyst review of sample alerts 5. Adjust thresholds based on FP feedback	Staging alert volume within 20% of production No CRITICAL alerts on known-good traffic Peer sign-off documented	Switch staging to production rules Redeploy previous rule version
Production	1. Schedule change window with NOC 2. Deploy rules to analyzer containers 3. Monitor alert queue for 1 hour post-deploy 4. Verify expected rule count loaded 5. Update rule documentation version	Alert rate stable (+/- 20% of pre-deployment) No analyzer errors in logs Dashboard shows updated rule version	Instant rollback: reload previous rules file Auto-rollback if error rate >50% increase

7.3 Performance Tuning & Capacity Planning

As Djazzy's subscriber base grows and traffic patterns evolve, the SS7 module requires periodic capacity adjustments. Proactive capacity planning prevents performance degradation during traffic spikes or seasonal events (Ramadan, holidays, promotions). Monitor leading indicators to predict capacity needs before they become critical.

Metric	Current Threshold	Warning At	Critical At	Scaling Action
Kafka consumer lag (avg)	<1000 messages	>5000	>20000	Add consumer instances
Collector CPU utilization	<60%	>75%	>90%	Scale horizontally or optimize decode
Analyzer processing latency	<500ms (p99)	>1s	>3s	Add analyzer instances, tune rules
Elasticsearch indexing latency	<2s	>5s	>10s	Add index nodes, reduce refresh interval
PCAP disk utilization	<70%	>80%	>90%	Expand storage, accelerate archival
Memory usage per container	<70% of limit	>85%	>95%	Increase memory limit, check for leaks
Message buffer saturation	<50%	>70%	>90%	Increase buffer size, speed up consumers

8. Quick Reference: Health Checks & Diagnostics

This chapter provides quick reference material for common diagnostic and troubleshooting tasks. Print or bookmark this section for rapid access during incident response when time pressure prevents consulting more detailed chapters. Commands assume execution from the SOC operations jump host with proper SSH keys and environment variables configured.

8.1 Essential Diagnostic Commands

Check	Command	Healthy Output Example	Common Issues
Collector health	curl -s http://ss7-collector:7000/health python3 -m json.tool	{\"status\":\"healthy\",\"messages_processed\":\"1234567\",\"spins\":\"86400\"}	Collector down, message backlog
Analyzer health	curl -s http://ss7-analyzer:7001/health python3 -m json.tool	{\"status\":\"healthy\",\"rules_loaded\":\"42\",\"ruleset_version\":\"2.0.0\"}	Rules not reloaded, config not missing
Kafka topics	kafka-topics.sh --bootstrap-server kafka-1:9092 --list	ss7-raw-events, ss7-alerts, diameter-events	Topics missing: create failed
Consumer lag	kafka-consumer-groups.sh --bootstrap-server kafka-1:9092 --list	ss7-raw-events ss7-analyzer	High lag: consumers slow/blocked
ES cluster health	curl -s es-master:9200/_cluster/health?pretty grep status	\"status\" : \"green\"	yellow/red: shard issues
ES index size	curl -s es-master:9200/_cat/indices/ss7-*?v	ss7-raw-2025.08.01 15GB 2000000	Index too large: adjust retention
Disk space (collector)	ssh collector \"df -h /var/capture/ss7\"	/dev/nvme1n1 2.0T 800G 1.2T 40% /dev/sda1 400G 300G 100G 75%	Disk full, need expansion
Docker container status	docker compose -f docker-compose.prod.yml ps ss7	ss7-collector Up (healthy) 0.0.0.0:2901->2901	Container crashed, restart loop
TheHive connectivity	curl -s -H \"X-API-Key: \$THEHIVE_KEY\" thehive:9000/api/ssl/analyze python3 -m json.tool	{\"ok\":true,\"analysis_id\":\"12345\"}	Unauthorized: expired/invalid key

8.2 Common Issues Quick Fixes

Symptom	Quick Fix	If Fix Fails
No alerts despite traffic	Check if analyzer rules loaded: curl analyzer:7001/health. Restart analyzer if rules missing.	Check analyzer permissions, verify ES index exists
High false positive rate	Temporarily increase threshold in rule YAML, redeploy, monitor for 24 hours.	Engage Rules Engineering for fundamental fix
Kafka consumer stuck	Restart consumer group: kafka-consumer-groups --reset-offsets --zookeeper-quorum=zk1:2181,zk2:2181,zk3:2181 --command-prefix kafka-topics.sh --bootstrap-server kafka-1:9092 --group ss7-raw-events --topic ss7-raw-events --auto-offset-reset=earliest --dry-run	Check for partition imbalance, rebalance cluster
TheHive case creation failing	Verify API key validity, check rate limit status in TheHive admin console.	Open dead-letter, implement batching
PCAP rotation failing	Check disk space, permissions on /var/capture. Manually rotate and archive old PCAPs.	Expand storage, check archive job configuration
Grafana datasource error	Test connection in Grafana DataSource settings. Refresh credentials if expired.	Check ES/Kafka service health, network connectivity

8.3 Emergency Contacts Reference

Role	Primary Contact	Backup Contact	Escalation Path	Availability
SOC Manager (shift)	On-call PagerDuty schedule	Deputy SOC Manager (phone list)	CISO after hours	24/7 (primary 8am-8pm)
NOC Engineer (infrastructure)	NOC PagerDuty schedule	NOC Manager (phone list)	IT Director after hours	24/7
Network Security (STP/firewall)	NetSec team email + Slack channel	Network Director (email)	CTO for emergency changes	Business hours + on-call weekend

Role	Primary Contact	Backup Contact	Escalation Path	Availability
Privacy Officer	privacy@djezzy.dz (email)	DPO Deputy (email)	Legal Director (email)	Business hours, 4hr response SLA
Fraud Team	fraud-team@djezzy.dz (email)	Fraud Manager (phone)	CFO office for executive issues	Business hours + IRSF on-call
Legal/Law Enforcement Liaison	legal-security@djezzy.dz (email)	General Counsel (phone)	External counsel for litigation	Business hours, 2hr urgent SLA
Vendor Support (SS7 tools)	Contract support portal + account manager	Technical account manager	Vendor escalation matrix	Per contract SLA

Appendix A: Checklist Summary

Checklist	Frequency	Location	Sign-off Required
Shift Start Health Verification	Each shift	Section 2.1	Self (documented in log)
Overnight Alert Review	Each shift	Section 2.1	Self (documented in log)
Baseline Metrics Capture	Each shift	Section 2.1	Self (documented in log)
Dashboard Anomaly Check	Every 15 min	Section 3.1-3.2	Self (alert if anomalous)
Handover Preparation	End of shift	Section 2.3	Outgoing analyst
Handover Acceptance	Start of shift	Section 2.3	Incoming analyst
Weekly Rule Effectiveness Review	Tuesday	Section 7.2	Rules Engineer + SOC Lead
Monthly Backup Verification	15th of month	Section 7.1	SOC Engineer + Backup Admin
Quarterly Capacity Planning Review	Quarter start	Section 7.3	SOC Architect + Capacity Team
Bi-Annual DR Test	Jan / Jul	Section 7.1	Full SOC team (test report)
Post-Incident Review (per major incident)	Within 1 week of closure	Section 4.x (runbook closure)	SOC Lead + involved analysts

Appendix B: Key Metrics Definitions

Metric	Definition	Calculation	Target Range
EPS (Events Per Second)	Rate of SS7 messages processed by collector	$\text{Collected messages} / \text{second}$	35,000 - 50,000
MTTD (Mean Time To Detect)	Average time from attack start to alert generation	$\text{Sum}(\text{detection_time} - \text{attack_start}) / \text{N alerts}$	60 seconds (automated), <15 min (behavioral)
MTTR (Mean Time To Respond)	Average time from alert to initial containment	$\text{Sum}(\text{response_time} - \text{alert_time}) / \text{N alerts}$	5 minutes (CRITICAL), <30 min (HIGH)
False Positive Rate	Percentage of alerts dismissed as noise	$\text{Dismissed alerts} / \text{Total alerts} * 100$	<20% (per rule), <15% (overall)
Consumer Lag	Messages in Kafka awaiting consumption	$\text{Topic_end_offset} - \text{consumer_offset}$	<10,000 (sustained)
Detection Coverage	Percentage of known attack patterns/rules covered	$\text{Rules_covering_attack_types} / \text{Total known types}$	100% (per rule, 30-day rolling)
Rule Precision	Percentage of alerts that are true positives	$\text{True_positive_alerts} / \text{Total alerts} * 100$	98% (per rule, 30-day rolling)
Revenue Saved (IRSF)	Estimated fraud loss prevented by blocking	$\text{Blocked calls} * \text{avg_premium_rate} * \text{avg_duration}$	Target monthly, aim >\$100K/month

Appendix C: Document Control

Property	Value
Document Title	Djezzy SOC Platform - SS7 Security Module Operations Manual
Document Owner	SOC Operations Manager
Version	2.0
Effective Date	2026-08-03
Review Cycle	Quarterly (or after significant incident)
Next Review Date	(Calculated from effective date + 3 months)
Approval Authority	CISO + SOC Director (joint sign-off)
Distribution	Internal - SOC, NOC, Security Engineering, Compliance
Classification	Internal Operations - Confidential